Summary:

SMI3 – Business Validation and Model Refinement

- Rakesh and the modeling team presented the completed multi-phase business validation of SmartMI 3. Underwriters reviewed 2,400 targeted files over approximately three and a half weeks, with an overall agreement rate of 82%.
- The model was simplified from 18 to 12 features to improve consistency, transparency and explainability while maintaining predictive performance above the minimum AUC threshold.

Discussion Summary

- Disagreements from the underwriter review informed model refinements. TDS and GDS treatment was simplified to capture non-housing debt; loan-purpose categories were reduced where they did not add material predictive value.
- The team noted that refinance-related risk should be reassessed if refinance products are reintroduced; manual review may be appropriate where historical data are limited.
- A small reduction in AUC was accepted in exchange for more intuitive, stable and explainable model behavior. Back-testing indicated broadly unchanged predictive performance.

SMI3 – Model Stability, Technical Validation and Usage Controls

- The modeling team described binning, monotone constraints and hyperparameter optimization to address counterintuitive risk relationships and improve model stability.
- Technical validation recorded 27 findings, of which seven were resolved; remaining findings primarily concern supporting documentation, rationale, and Jira acceptance criteria.
- The team clarified the proposed production usage boundary: SmartMI 3 is intended for file-level risk ranking under the specified 60th-percentile economic scenario only, not for loss forecasting; scenario changes require reassessment and recalibration.

Discussion Summary

- Monotone constraints were applied to preserve expected relationships for credit score, bankruptcy and other risk drivers. Hyperparameter optimization reduced observed instability from approximately 30% to 4%.
- Independent optimization produced similar results; swap-set testing showed more than 99.8% of files moved within 5%. Tests changing delinquency labels in approximately 460,000 records shifted losses by about 6%, while independent testing suggested approximately 13% loss uncertainty.
- Changes to the economic scenario, including HPI depreciation assumptions, require reassessment, validation and calibration. The restriction is on permitted use rather than a defect in the trained model.
- The team has completed more than 50 incremental model iterations and continues final testing and documentation.

Action Item

- Rakesh and the Technical Validation Team to complete outstanding experiments and document rationales and acceptance criteria for Jira findings.
- Business Team to document approved scenario usage, file-level ranking restrictions and recalibration requirements.

SMI3 – LGD Formula and Credit Score Updates

- The team presented LGD formula refinements intended to improve property-level risk discrimination, including more granular distress factors, property-value-based expenses and condo-specific treatment.
- SmartMI 3 uses TransUnion TrueVision as the primary credit score and FICO 10 as backup, and uses the maximum rather than average borrower score for multiple-borrower files.

Discussion Summary

- Distress factors now account for condo status and property age in addition to geography and urban-rural distinctions. Insurance expenses are calculated as a share of property value; maintenance includes a fixed baseline and variable component.
- Condo maintenance treatment incorporates condo fees without double counting. File-level contract interest rates were removed; variable-rate mortgages retain a higher risk contribution than fixed-rate mortgages, with the PRM contribution reduced to approximately 5% from 10%.
- The revised credit-score approach improves alignment of risk ranking with capital treatment and operational decisions.

SMI3 – Swap Set Results and Atlantic Regional Impact

- Comparison of SmartMI 2 and SmartMI 3 showed broadly similar overall results and improved risk separation; fewer than 9% of test files showed significant movement.
- The Atlantic region showed a larger change in modeled loss ratio than other regions. The team completed back-testing and underwriter review and found the revised ranking consistent with expected risk.

Discussion Summary

- Loans leaving decile one had an average credit score of 725, compared with 699 for those entering; the incoming group had higher proportions of LTV above 90% and single borrowers.
- The Atlantic impact reflects differences in non-housing debt, expenses, rural mix and property characteristics. Atlantic represents approximately 6–7% of new insured business.
- The committee requested an urban-rural breakdown of Atlantic results. Regional impacts will be considered when setting automated decision thresholds.

Action Item

- The team to provide the Atlantic loss-ratio grid separately for urban and rural segments to the committee.

SMI3 – LEE Plus Pricing Assessment

- Jacklyn presented the adaptation of SmartMI 3 outputs to the existing LEE Plus pricing infrastructure and reported little overall premium impact, with capital remaining the primary pricing driver.
- Testing of January–June 2026 applications at an 8.5% IRR target showed approximately 96% of loans had premium differences within plus or minus one basis point and 80% had no premium change.

Discussion Summary

- A 120-period loss pattern was developed from the 2025 LEE Plus population using double-exponential smoothing because SmartMI 3 does not produce the monthly cash flows used by the SmartMI 2 pricing infrastructure.
- SmartMI 3 produced higher modeled losses at lower LTVs because of updated distress factors, while maintaining consistent risk ranking.
- At a flat 13% return target, SmartMI 2 averaged approximately 34.8 basis points; using SmartMI 3 losses and TrueVision reduced average premium by approximately 0.2 pips.

SMI3 – Implementation Timeline and Final Committee Decision

- At the end of the discussion, the committee approved the business validation subject to completion of the final technical validation tests and delivery of the final business-validation documentation. This was approval of the validation work, not final approval of the model for deployment.
- The team outlined plans for final MSC approval, production parallel testing, disaster-recovery deployment and a target release on December 11, 2026, with January as the contingency if unexpected rework delays implementation.

Discussion Summary

- Training-code finalization was targeted for October 21 and inference-code finalization for October 28, with a project check-in on October 28.
- The team plans to return to MSC on November 6 for final technical validation and readiness approvals; version 51 is targeted for production parallel testing around that time.
- The model is already running behind the scenes in production. The approved version is planned for parallel testing and deployment in a separate AWS region for disaster recovery.
- Automated decision thresholds will be developed after validation, initially using a do-no-harm baseline to preserve key customer and portfolio metrics.

Action Item

- Technical Team and Silverio to finalize inference code and prepare production parallel testing and disaster-recovery deployment.
- The team to return to MSC with completed validation, final documentation and model iteration for formal approval.
- Rakesh to request committee members' email approval of the circulated meeting minutes.

Action Items Summary:

Action Item	Owner	Timeline
Provide final business-validation documentation, including approved scope and limitations.	Eugene	Next MSC meeting
Complete outstanding technical-validation experiments and Jira documentation.	Rakesh	Before final approval
Provide Atlantic loss-ratio grid by urban and rural segment.	The team	Not specified

Document 60th-percentile scenario restriction and reassessment requirements.	Rakesh / Business Team	Before production use
Finalize inference code and prepare parallel testing and disaster recovery.	Santhosh/ Silverio	Oct. 28 target / before release
Return to MSC for final validation and readiness approval.	The team	Nov. 6, 2026
Obtain email approval of circulated meeting minutes.	Rakesh	Not specified